

Integration Testing

Validation of interactions among components, services, databases, queues and external systems.

What is Integration Testing?

Integration Testing verifies whether components, services, APIs, databases, queues or different systems exchange information and work together correctly. A module can work in isolation and still fail when it depends on another system.

What to validate

Contracts, payloads, formats, status codes, authentication, authorization, timeouts, retries, idempotency, mappings, transformations, persistence, event ordering and error handling.

Practical example

An e-commerce platform sends an order to the ERP. Validate order creation, mapped fields, values, taxes, address, identifiers, ERP response, status update and behavior when the ERP is unavailable.

Negative scenarios

400/401/403/404/409/500 response, timeout, incomplete payload, unexpected field, duplication, out-of-order message, unavailable dependency and retry. Integrations also need to be tested when something goes wrong.

Mocks, stubs and real environments

Mocks and stubs help isolate behavior and simulate difficult errors; integrated environments reveal real configuration, network and contract problems. Know the limitation of each approach and combine them according to risk.

Observability

Logs, traces, correlation IDs, metrics and queues help locate the failure boundary. Without observability, an integration error can look like a user interface defect.

Data and consistency

Confirm not only the immediate response, but the final state: persisted data, duplicates, consistency between systems and eventual consistency when applicable.

Quick checklist

Contract; authentication; payload; mapping; success; errors; timeout; retry; idempotency; persistence; logs; traceability; final consistency.



QA Mindset

Do not validate only “A called B”. Validate whether the right information arrived, was interpreted correctly and produced the expected effect end to end across boundaries.